AI.204

Autonomous Systems

COURSE OVERVIEW

Systems that operate without continuous human input: autonomy levels, decision making, and self-management.

1

SESSION

60 MINUTES

ORIENTATION: TOWARD AUTONOMY

KEY TOPICS

- Levels of autonomy
- Sense–plan–act autonomy
- Benefits and risks
- Autonomy in AI and robotics

TEACHING FOCUS

Define what it means for a machine to act alone.

KEY TAKEAWAY Autonomy is delegated decision-making.

2

SESSION

60 MINUTES

AUTONOMY ARCHITECTURE

KEY TOPICS

- Hierarchical vs. reactive
- Behavior trees
- Finite state machines
- Supervisory control

TEACHING FOCUS

Structure autonomous decision-making.

KEY TAKEAWAY Architecture determines how autonomously a system a

3

SESSION

60 MINUTES

MISSION & TASK PLANNING

KEY TOPICS

- Mission specification
- Task decomposition
- Contingency handling
- Replanning

TEACHING FOCUS

Turn goals into executable missions.

▲ **KEY TAKEAWAY** Planning under autonomy must anticipate surprises.

4

SESSION

60 MINUTES

DECISION MAKING UNDER UNCERTAINTY

KEY TOPICS

- POMDPs introduction
- Belief states
- Policy and value
- Online decision making

TEACHING FOCUS

Choose actions when the world is unclear.

KEY TAKEAWAY Uncertainty is the rule, not the exception.

5

SESSION

60 MINUTES

SAFETY IN AUTONOMY

KEY TOPICS

- Safe autonomy envelopes
- Monitoring and fallback
- Human-in-the-loop
- Validation of autonomous systems

TEACHING FOCUS

Keep autonomous systems within safe bounds.

KEY TAKEAWAY Autonomy without safety is dangerous.

6

SESSION

60 MINUTES

AUTONOMOUS SYSTEMS & REVIEW

KEY TOPICS

- Case studies: self-driving, drones
- Course recap and assessment
- Toward mobile robotics (AI.207)
- Autonomy demo

TEACHING FOCUS

Operate a system with minimal supervision.

KEY TAKEAWAY True autonomy earns human trust.